The Magic of Exponentials: Growth, Decay, and Compounding

The Meenakshi School — Applied Mathematics & Science Series

Executive Overview

This lesson explores how the exponential function $y = a \cdot b^x$ governs fundamental phenomena across the Universe—from ancient algorithmic puzzles and subatomic physics to modern finance and cosmology.

1. Classical Growth: The Wheat and Chessboard Problem (Chaturanga)

- **The Premise:** The inventor of chess requested a seemingly modest reward from the King: 1 grain of rice on the first square, 2 on the second, 4 on the third, doubling on each subsequent square across the 64-square board.
- **The Mathematics:**

$$\text{Total Grains} = \sum_{k=0}^{63} 2^k = 2^{64} - 1 = 18,446,744,073,709,551,615$$

- **The Reality Check:** Over 18.4 quintillion grains of rice—equivalent to nearly **3,000 years of global agricultural production**.

2. Exponential Decay: Medical Physics & Nuclear Tracers

- **The Concept:** While growth multiplies by $b > 1$, decay scales down by $0 < b < 1$. Nuclear medicine uses short-lived radioactive isotopes like **Technetium-99m ($^{99\mathrm{m}}\mathrm{Tc}$)** for SPECT imaging because they clear quickly from the human body.

Radioactive tracers are special chemical compounds made by attaching a radioactive atom like Technetium-99m to a natural body molecule like sugar or protein. Once inside the body through an injection, drink, or breath, they follow normal biological paths, travel to specific active tissues, and emit safe, detectable energy signals that external scanners turn into detailed medical images.

- **What are Tracers and How do they Move and Work**
  - **Delivery:** A patient receives the tracer via an injection, swallows it, or inhales it into the lungs.
  - **Travel:** The tracer mixes with blood or fluid and moves to target organs or high-energy cells.
  - **Cell Action:** Diseased or fast-working cells—like growing tumors or healing bone—absorb more or less of the tracer than healthy cells do.
  - **Signal Emission:** As the unstable atoms break down over time, they give off tiny rays of energy (gamma rays or positrons) that pass safely out through the skin.
- **Detection and Removal**
  - **Scanning:** Special external tools like a gamma camera or PET scan catch the escaping energy signals.
  - **Mapping:** Computers turn those signals into a map showing "hot spots" of high activity or "cold spots" of low activity.
  - **Leaving the Body:** The radioactive material loses its power quickly and leaves the body naturally through urine or sweat within a few hours or days.
- **The Decay Equation:**

$$N(t) = N_0 \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$$

  - N_0: Initial administered dose
  - $t_{1/2}$: Half-life (6 hours for $^{99\mathrm{m}}\mathrm{Tc}$)
  - t: Elapsed time in hours
- **Class Exercise:**
  - *Question:* If a patient receives a dose at 8:00 AM, how much remains in the body at 8:00 AM the following day (24 hours later)?

- *Calculation:* $t/t_{1/2} = 24/6 = 4 \text{ half-lives.}$
- *Result:* $\left(\frac{1}{2}\right)^4 = \frac{1}{16} = 6.25\%$ remaining.

3. Financial Compounding: The 10 Paise Doubling Effect

- **The Scenario:** Compare making a single large deposit versus doubling a tiny initial amount of $10 \text{ paise } (0.10$ ₹) every day for 30 days.
- **Day 30 Single-Day Value:**

$$\text{Value}_{\text{Day 30}} = 0.10 \times 2^{29} = 0.10 \times 536,870,912 = 53,687,091.20 \text{ ₹}$$

- **Cumulative 30-Day Account Balance:**

$$\text{Total Balance} = \sum_{k=0}^{29} (0.10 \times 2^k) = 0.10 \times (2^{30} - 1) = 107,374,182.30 \text{ ₹}$$

- **Takeaway:** Over 107 Million ₹ (or 10.74 Crores) accumulated from a starting balance of just 10 paise and doubling it for 30 days!

4. Universal Comparative Summary

Application Domain	Mathematical Model	Base Behavior (b)	Real-World Impact
Ancient Algorithmic Growth	$y = 2^x$	$b = 2 > 1$	Outstrips global planetary supply $(18.4 \text{ quintillion})$
Nuclear Medicine (Decay)	$y = \left(\frac{1}{2}\right)^{x/6}$	$b = 0.5 < 1$	Clears 93.75% of tracer radiation within 24 hours
Financial Compounding	$y = 0.10 \cdot (2)^x$	$b = 2 > 1$	Converts 10 paise into $>$ 10.7 Crores in 30 days